

New Jersey's New Dam Disclosure Law:

From Disclosure to Better Dam Stewardship

Engineering Perspectives Following Enactment of New Jersey's Dam-Disclosure Law

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August 14, 2026

Highlights

- **The law reveals; it does not create.** A deteriorating dam is a responsibility whether or not it is disclosed. Disclosure makes it visible before a sale.
- **It helps buyers and current owners.** Buyers learn what they are taking on. Owners are prompted toward inspection and preventive maintenance.
- **The adopted law now explicitly defines “dam.”** The introduced bill did not define the term. The final law uses a 5-ft water-level-rise criterion that closely tracks New Jersey's existing dam-safety definition and should not be confused with the separate regulatory measurement of “height-of-dam.”
- **The Pinelands exception is narrow and combined.** A dam there is excluded only if it raises the waters of a river or stream less than 8 ft above the ground surface, drains less than 1 square mile, and creates a water surface of less than 100 acres. A dam meeting the general 5-ft criterion can therefore still qualify for the exception.
- **Hazard class is not condition.** Classes I–III primarily describe the potential consequences of failure, while Class IV combines low-hazard consequences with explicit small-dam criteria. A high-hazard dam can be well maintained; a low-hazard dam can be in poor condition.
- **The inventory has boundaries.** The State Auditor counted 1,728 Class I–III dams as of April 2024, about 60% privately owned. Class IV and other qualifying small dams may meet the disclosure definition yet sit outside that count. No authoritative statewide Class IV count was identified.
- **Inventories can have gaps.** After the 2004 storms, some failed dams had not been recorded in the State database. Modern LiDAR, GIS, remote sensing, and AI could help identify candidate structures for further review and field verification.
- **A dam can look fine and still have serious problems.** Seepage, internal erosion, gate and outlet reliability, and spillway capacity often cannot be judged by eye. Professional assessment matters.
- **Costs vary enormously.** Simple disclosure may cost little. Studies, repairs, and rehabilitation scale up quickly. A USGS analysis of 668 removals found median removal costs of about \$157k, \$823k, and \$6.2M for small, medium, and large dams.

- **Disclosure does not solve funding.** Owners of deteriorating private dams may still face large costs. Financing and cost-sharing deserve attention, especially where a private dam protects public roads, utilities, or people.
- **Dams are not only liabilities.** Depending on the site, they can provide water supply, recreation, habitat, historical, cultural, and aesthetic value. Decisions should therefore extend beyond a simple “keep or remove” choice.

Overview

On August 14, 2026, NJ.com published an article on New Jersey’s newly enacted dam-disclosure law, which requires sellers of residential property to disclose specified information when a dam is located wholly or partly on the property. The article describes the experiences of New Jersey homeowners who unexpectedly encountered substantial responsibilities and costs associated with dams and discusses what the new law may mean for future buyers and sellers.

I was interviewed for and quoted in the article as a professor of civil and environmental engineering at Rutgers University. Among the points I discussed were the value of disclosure to prospective buyers, the importance of professional dam assessments rather than relying on visual appearance alone, and the additional stresses that extreme rainfall can place on aging dams. The article highlighted these points in its discussion of the new law.

This reflection complements my earlier March 2026 brief, *Dam Disclosure*, prepared while the legislation was still being considered. That brief focused particularly on hidden dams, homeowner responsibilities, field identification, and modern methods for screening possible dams and embankments. The present reflection shifts the discussion from whether disclosure is desirable to **how the enacted law can be implemented effectively**.

1. Why Dam Disclosure Matters

The problem addressed by the new law is straightforward but consequential: a person may purchase property without fully appreciating that a pond, earthen embankment, or other landscape feature includes a dam—and that ownership of the property may bring responsibilities extending well beyond ordinary landscaping.

Depending on the dam and its classification and condition, ownership can involve inspection, maintenance, engineering evaluation, repair, rehabilitation, emergency planning, or eventually removal.

The NJ.com article illustrates two somewhat different information problems. One homeowner reported discovering the dam only after purchasing the property. Another knew that a dam was present but said he had not understood the seriousness of its longstanding condition.

These cases illustrate why disclosure should involve more than simply checking a box saying “**there is a dam.**”

A prospective buyer should, where applicable, be able to learn:

- that a dam exists;
- its hazard classification;
- available inspection information;
- known deficiencies;
- applicable owner responsibilities; and
- potentially significant future maintenance or repair obligations.

The enacted legislation specifically requires the property condition disclosure statement to address whether a dam is located wholly or partly on the property, its hazard classification, applicable inspection results, and State-law obligations that a purchaser would assume.

The disclosure law therefore helps reduce an important information imbalance between sellers and prospective buyers.

2. Disclosure Can Also Benefit Current Dam Owners

Although disclosure is naturally viewed as consumer protection for buyers, I believe it can also benefit **current dam owners**.

The process can prompt an owner to determine whether the structure is listed by the New Jersey Department of Environmental Protection (NJDEP), understand its classification, review available inspection records, and become more aware of inspection and maintenance responsibilities.

That awareness can encourage preventive maintenance.

A relatively minor erosion problem, obstructed outlet, animal burrow, excessive vegetation, deteriorating concrete, developing seepage condition, or mechanical problem may be much less expensive to address when identified early than after it develops into a major repair or safety problem.

The potential progression is:

Disclosure → awareness → inspection and maintenance → earlier identification of deficiencies → timely repair → potentially lower long-term costs and reduced failure risk.

Thus, disclosure should not be viewed only as an obligation imposed when a property is sold. It can contribute to better stewardship of the dam while the current owner still owns it.

3. What Exactly Counts as a “Dam”? Understanding the Five-Foot Threshold

One improvement in the enacted legislation is greater clarity concerning what constitutes a dam. **The introduced bill required disclosure of a “dam” but did not itself define the term. The amended and enacted legislation added an explicit definition that closely follows the definition already used under New Jersey’s existing Dam Safety Standards, N.J.A.C. 7:20. Thus, the 5-ft water-level-rise criterion is not new; what is new here is its explicit inclusion in the dam-disclosure law.**

The 5-ft criterion should not simply be described as “a dam five feet high.” The statutory language concerns the amount by which an artificial barrier raises the water level.

The law defines a dam as an artificial dike, levee, or other barrier, together with appurtenant works, constructed to impound water permanently or temporarily, that raises the water level **5 ft or more above the usual mean low-water height**, measured from the downstream toe of the dam to the emergency-spillway crest or, if there is no emergency spillway, to the top of the dam.

This is important because N.J.A.C. 7:20-1.2 separately defines “**height-of-dam.**”

N.J.A.C. 7:20-1.2 separately defines “height-of-dam” as the vertical dimension from the lowest point in the streambed or ground surface at the downstream toe of the dam to the elevation of the top of the dam (without camber). The regulation also separately defines “toe-of-dam” as the junction of the downstream face of the dam with the ground surface or the invert of the outlet pipe, whichever is the lowest point.

The 5-ft water-level-rise criterion and the separately defined height-of-dam, therefore, should not be casually interchanged. The relevant regulatory definitions can be summarized as follows:

Question	Applicable regulatory measurement
Does the structure meet the general definition of a “dam”?	The structure must raise the water level 5 ft or more above the usual mean low-water height, measured from the downstream toe-of-dam to the emergency-spillway crest, or to the top-of-dam if there is no emergency spillway.
What is the “height-of-dam”?	The vertical dimension from the lowest point in the streambed or ground surface at the downstream toe of the dam to the top of dam.
What is the “toe-of-dam”?	The junction of the downstream face of the dam with the ground surface or the invert of the outlet pipe, whichever is the lowest point.

This distinction also explains why a homeowner should not simply look at an embankment and decide that it “looks about four feet high” and therefore is outside the law.

For a borderline structure, a reasonable progression would be:

Check NJDEP information → contact NJDEP if necessary → obtain professional measurement or engineering assistance where uncertainty remains → document the determination.

Indeed, the enacted legislation specifically provides that sellers and purchasers may contact the NJDEP Bureau of Dam Safety for assistance in determining whether a dam is subject to the Safe Dam Act or associated regulations.

Clear diagrams and examples from the State explaining the measurement could make implementation easier for homeowners, real-estate professionals, and engineers.

The law also contains a special provision for certain dams within the Pinelands Area.

4. A Special Pinelands Exception

The new law includes a special exception for certain dams located within the **Pinelands Area**.

Under the statutory language, the term “dam” does not include a dam in the designated Pinelands Area when **all three** of the following conditions are satisfied:

- the dam raises the waters of a river or stream **less than 8 ft above the surface of the ground**;
- the drainage area above the dam is **less than 1 square mile**; and
- the water surface created by the dam is **less than 100 acres**.

This provision is important because the general definition uses a **water-level rise of 5 ft or more**, whereas the Pinelands exception uses a combination of water-level rise, drainage area, and water-surface area.

Consequently, a structure in the Pinelands can meet the general statewide 5-ft criterion but nevertheless be excluded from the definition under the special Pinelands provision. For example, a structure that raises the waters of a river or stream by 6 or 7 ft could fall within the exception if its drainage area is less than 1 square mile and the resulting water surface is less than 100 acres.

Conversely, satisfying only one or two of these conditions is not sufficient for the exception. The statutory language connects all three.

For purposes of understanding the disclosure law:

The law generally defines a dam using a water-level rise of 5 ft or more, while providing a special Pinelands exception based on three combined criteria: water-level rise, upstream drainage area, and the water-surface area created by the dam.

This is another reason that a property owner should not determine whether a structure is subject to disclosure merely by estimating its apparent physical height.

5. Hazard Classification Is Not the Same as Dam Condition

New Jersey classifies dams primarily according to the **potential consequences if the dam fails**, not according to whether the dam is presently in good or bad condition.

This distinction is fundamental.

Class I — High Hazard Potential

Failure may result in probable loss of life and/or extensive property damage.

There is no single height, storage-volume, or drainage-area threshold that makes a dam Class I. The classification is principally consequence-based.

Class II — Significant Hazard Potential

Failure may result in significant property damage, although loss of life is not envisioned.

Again, no single numerical dam size defines Class II.

Class III — Low Hazard Potential

Failure is not expected to result in loss of life and/or significant property damage.

Class III likewise is primarily consequence-based rather than defined by one dimensional threshold.

Class IV — Small-Dam Low-Hazard Potential

Class IV differs because it combines low-hazard consequences with explicit small-dam criteria. Under N.J.A.C. 7:20-1.8, Class IV criteria include:

- height-of-dam **less than 15 ft**;
- potential impoundment **less than 15 acre-ft**;
- drainage area of **150 acres or less**;
- low-hazard potential; and
- specified spillway-capacity requirements.

The central distinction is:

Hazard classification describes the consequences of failure. Condition describes how well or poorly the dam is performing today.

A Class I high-hazard dam can be very well maintained and in excellent condition. Conversely, a low-hazard dam can be in poor physical condition.

6. A Dam Can Look Fine and Still Have Serious Problems

This was one of the issues I emphasized during my NJ.com interview, and the published article noted the importance of professional assessment rather than relying solely on appearance.

A visual walk-around is valuable. An observer may notice:

- erosion or gullies;
- cracks or settlement;
- displaced riprap;
- excessive trees and vegetation;
- animal burrows;
- deteriorated concrete;
- obvious seepage;
- clogged outlets;
- damaged spillways; or
- other visible abnormalities.

But visual appearance alone cannot establish that a dam is safe.

Seepage and internal erosion

This is particularly important for earthen dams.

Water naturally moves through and beneath an earth embankment. The engineering question is whether that seepage is being safely controlled.

Internal erosion—often referred to as **pip**ing—can progressively remove soil particles from within the embankment or foundation. Early development may be difficult for a homeowner to recognize. Signs can be subtle, intermittent, hidden by vegetation, or located where an untrained observer would not appreciate their significance.

By the time muddy seepage, sinkholes, settlement, or other obvious symptoms appear, the problem may already be serious.

Gates and outlet works

Mechanical components present another example.

A gate may look perfectly normal but fail to operate when it is needed because of:

- corrosion;
- mechanical deterioration;
- sediment or debris;
- electrical failure;
- problems with the lifting mechanism; or
- long periods without full operational testing.

The mere presence of a gate therefore does not demonstrate that the gate can be fully opened during an emergency.

Spillway capacity

Likewise, a spillway can look structurally sound yet have insufficient capacity for an extreme flood.

The July 2026 **Liulan Reservoir failure in Guangxi, China**, which I have studied separately, provides a recent reminder. During an extraordinary rainfall and flood event, reported operational difficulties included gates that could not all be fully opened as intended, followed by overtopping and breach development.

The scale, design, regulatory setting, and circumstances differ substantially from the small private dams addressed by New Jersey's disclosure law. Nevertheless, the broader engineering lesson is relevant:

Dam safety depends not only on what can be seen from the outside, but also on internal seepage behavior, hydraulic capacity, mechanical reliability, maintenance history, and performance under unusual conditions.

For homeowners and prospective buyers:

A walk around a dam is valuable for identifying visible warning signs, but visual appearance alone cannot establish its condition or safety and is not a substitute for an appropriate professional dam-safety assessment.

7. What Is Actually in New Jersey's Dam Inventory?

The NJ.com article reports that New Jersey regulates **more than 1,700 dams**, including **232 classified as high hazard**, and that approximately 60% are privately owned.

The most detailed recent public breakdown I have identified comes from the **2025 New Jersey State Auditor's audit of the Bureau of Dam Safety**, using Bureau data as of April 2024.

At that time, the Bureau database contained **1,728 Class I–III dams**:

Classification	Number
Class I — High Hazard, Large	33
Class I — High Hazard, other	198
Total Class I	231
Class II — Significant Hazard	336
Class III — Low Hazard	1,161
Total Class I–III	1,728

The State Auditor also reported that **1,035 of the 1,728 dams, or approximately 60%, were privately owned**. The Auditor found the dam data sufficiently reliable for purposes of its report, while also identifying issues concerning inspections, enforcement, and maintenance of dam records.

These numbers need to be interpreted carefully.

The State Auditor’s 1,728-dam total specifically covers **Classes I, II, and III**. It should therefore not automatically be interpreted as a complete count of every physical structure in New Jersey that meets the definition of a “dam” used by the new disclosure law.

That distinction has become particularly relevant because the disclosure law creates an additional information need. The Bureau’s inventory has historically served primarily dam-safety regulation and management. For property disclosure, a homeowner may first need to answer a more basic question:

Does a structure on my property meet the law’s definition of a dam and therefore need to be disclosed?

This leads to an important question concerning Class IV small, low-hazard dams.

8. Class IV and the Inventory Question

Class IV dams deserve particular attention when considering the relationship between New Jersey’s dam-safety inventory and implementation of the new disclosure law.

As discussed in Section 3, the **5-ft water-level-rise criterion is not new to the disclosure law**. It is part of the general definition of a “dam” already used in New Jersey’s dam-safety framework and is now explicitly included in the enacted disclosure law. The introduced disclosure bill, by contrast, referred to a “dam” without defining the term.

Class IV is a **separate classification** for certain small, low-hazard dams under N.J.A.C. 7:20-1.8. Its criteria include:

- height-of-dam **less than 15 ft**;
- potential impoundment **less than 15 acre-ft**;
- drainage area of **150 acres or less**;
- low-hazard consequences; and
- specified spillway-capacity requirements.

An important technical distinction is that the **5-ft water-level-rise criterion used in the general definition of a dam and the Class IV 15-ft height-of-dam criterion are different measurements serving different regulatory purposes**. They should not be interpreted as the lower and upper limits of a single 5-to-15-ft physical dam-height range.

In practical terms, the first question is whether a structure meets the applicable regulatory definition of a **dam**, including the general 5-ft water-level-rise criterion. If it does, its separately determined **height-of-dam**, potential impoundment, drainage area, hazard consequences, and other applicable criteria become relevant to its classification. Thus, a structure can meet the general definition of a dam while also satisfying the separate size and hazard criteria for Class IV.

This distinction becomes important for implementation of the disclosure law. The law defines what constitutes a dam for purposes of disclosure; it does not limit disclosure only to dams appearing in the Class I–III inventory. Qualifying Class IV dams therefore may also be relevant to property disclosure even though they are not included in the State Auditor’s April 2024 total of **1,728 Class I–III dams**.

Some of these structures may not resemble what an ordinary homeowner imagines when hearing the word “dam.” They may include earthen embankments associated with old farm ponds, millponds, residential ponds, or other relatively small impoundments.

This leads to an important inventory question. I have not identified a current authoritative public statewide count of Class IV dams. Accordingly, the approximately **1,700-dam figure** commonly cited for New Jersey should not automatically be interpreted as the total number of structures potentially relevant to the new disclosure law.

The more useful implementation question is therefore:

How many Class IV and other dams meeting the disclosure-law definition are documented in State records readily available to property owners, and how many qualifying structures, if any, remain outside those records?

The publicly available information reviewed for this reflection does not provide a definitive answer.

This does **not** mean that Class IV dams are necessarily unknown to NJDEP. Individual structures may already be documented in agency, local, historical, inspection, engineering, permitting, or other records. The narrower point is that I have not identified a current, authoritative, publicly available **statewide Class IV count** comparable to the published Class I–III inventory.

Clarifying this inventory boundary—and making information on qualifying dams readily accessible—would help homeowners, prospective buyers, real-estate professionals, engineers, and regulators implement the new disclosure law more consistently. It would also help distinguish between two separate questions: **how many dams New Jersey regulates or has documented, and how many structures may meet the definition that triggers property disclosure.**

9. History Shows That Dam Inventories Can Have Gaps

This is not merely a theoretical concern.

Following the severe July 2004 storm in southern New Jersey, numerous dams failed or were damaged. A subsequent State task-force investigation examined the dam failures, emergency planning, inspection, enforcement, and related issues.

The report documented that seven of the failed dams had not previously been recorded in the Department’s database and therefore had no record of previous condition.

That historical experience does **not** establish the completeness of today’s database. New Jersey’s dam-safety program and inventory have evolved substantially during the two decades since.

It does, however, illustrate a broader point:

An official inventory should not automatically be assumed to contain every qualifying physical structure unless its completeness has been established.

Implementation of the disclosure law therefore provides another reason to continue improving New Jersey’s dam information.

10. Could Modern Technology Help Improve the Dam Inventory?

Yes.

Modern geospatial technologies provide opportunities that were not readily available when many dam inventories were originally assembled.

LiDAR and high-resolution digital elevation models can reveal:

- linear embankments;
- flat dam crests;
- upstream/downstream elevation differences;
- spillway channels;
- former stream channels; and
- subtle terrain features hidden beneath vegetation.

Aerial and satellite imagery can reveal:

- ponds and reservoirs;
- artificial shorelines;
- stream-spanning barriers;
- changes through time; and
- landscape features suggesting artificial impoundment.

GIS can combine these observations with stream networks, watershed boundaries, parcels, roads, existing inventories, and other spatial information.

Machine learning and artificial intelligence could then help screen large areas and identify candidate dams or dam-like structures for further review.

A possible workflow is:

Existing inventory → LiDAR/DEM screening → aerial/satellite imagery → stream and waterbody analysis → AI/ML candidate identification → historical records and imagery → prioritized field verification → inventory improvement

AI should not make the final regulatory determination.

Remote sensing and AI can identify candidates; engineering judgment, agency records, appropriate measurements, and field verification should confirm them.

This could be an appropriate area for collaboration among NJDEP, other government agencies, universities, local governments, and engineering professionals.

11. How Much Can Dam Ownership Cost?

The NJ.com article understandably highlights the potentially substantial financial burden faced by individual homeowners. It reports that, in one case, removing or taking apart the dam was estimated to cost nearly \$1 million, while repairing it could cost even more.

That is an important real-world example, but it should not be interpreted to mean that **dam repair normally costs \$1 million**.

Costs vary enormously.

The following figures are **illustrative planning ranges based on general engineering experience**. They are not derived from a statewide New Jersey cost database and are not NJDEP fee schedules, contractor quotations, or estimates for any particular property.

Potential need	Illustrative order of magnitude
Existing records / straightforward disclosure	Little or no significant additional cost
Limited survey or measurements	Hundreds to several thousand dollars
Initial engineering consultation/site review	~\$1,000–\$5,000+
Formal engineering inspection/report	~\$3,000–\$15,000+
Detailed hydraulic/geotechnical/survey/stability studies	~\$10,000–\$50,000+, sometimes more
Routine maintenance	Hundreds to several thousand dollars
Significant localized repair	~\$10,000–\$100,000+
Major rehabilitation	~\$100,000–\$500,000+
Complex rehabilitation	~\$500,000–\$1 million or more

Dam removal can also be expensive because it may involve engineering, permitting, water diversion, demolition, sediment management, stream restoration, bank stabilization, revegetation, and monitoring.

A USGS analysis compiled reported costs for **668 U.S. dam removals**. After adjustment to 2020 dollars, costs varied enormously—from approximately \$1,000 to \$268.8 million per dam. The median reported removal cost was approximately **\$157,000 for dams less than 5 m high**, \$823,000 for dams between 5 and 10 m, and \$6.2 million for dams higher than 10 m. Dam height, average discharge, and project complexity were among the principal cost drivers.

The important point is:

The disclosure law does not create a deteriorated spillway, failing outlet, unstable embankment, or expensive rehabilitation problem. It reveals an infrastructure responsibility that already exists.

12. Disclosure Does Not Solve the Funding Problem

The NJ.com article raises another important issue through comments from a dam owner: disclosure can protect future buyers, but it does not necessarily solve the financial problem faced by **current owners of deteriorating private dams**.

This distinction deserves attention.

Three related policy questions are:

Disclosure: Who needs to know about the dam?

Dam-safety regulation: What must be inspected, maintained, repaired, or otherwise addressed?

Funding: How can necessary work actually be paid for?

The new law primarily addresses the first.

The State Auditor reported that New Jersey has historically maintained bond-funded programs providing low-interest loans to dam owners for restoration and repair projects.

As the State implements the disclosure law, consideration of financing mechanisms, technical assistance, cost sharing, loans, grants, or other programs may become increasingly important—particularly where a privately owned dam also protects downstream people, roads, utilities, or other public interests.

13. Liability and Insurance

Dam ownership can also create legal and financial exposure if failure results in injury or damage to neighboring or downstream property.

Dam failures in the United States have resulted in litigation. Whether an owner is ultimately legally liable depends upon the particular circumstances and applicable law; being sued following a failure is not the same as being found liable.

At the same time, I am not aware of a general New Jersey requirement that every private dam owner purchase **dam-specific liability insurance simply because the property contains a dam.**

These are different issues:

No general dam-specific insurance mandate does not mean no potential financial exposure.

Owners should understand what their homeowner, general-liability, association, or other insurance does and does not cover and obtain appropriate professional insurance or legal advice where necessary.

Insurance also does not substitute for proper inspection, maintenance, repair, and emergency preparedness.

Interestingly, following the 2004 failures, the State task force recommended examining whether insurance covering liability and property damage could be made available to dam owners.

14. Extreme Weather Makes Dam Awareness More Important

The NJ.com article also reported my observation that more frequent large storms can place additional strain on dams that were not designed for such conditions and may increase inspection and maintenance needs.

The engineering relationship is broader than rainfall alone.

Extreme rainfall can increase watershed inflow and hydraulic loading on reservoirs, spillways, outlets, and embankments.

For earthen dams, extreme events can contribute to:

- overtopping;
- erosion;
- increased seepage;
- internal erosion;
- slope saturation and instability;
- spillway stress; and
- outlet problems.

Risk depends on the interaction of rainfall magnitude and duration, antecedent wetness, reservoir level, available storage, watershed response, spillway capacity, outlet reliability, dam condition, and downstream consequences.

Older dams warrant particular attention because the hydrologic assumptions used when they were designed or constructed may differ from present or future conditions.

Extreme rainfall makes knowing what you own even more important.

Climate and hydrologic changes are not themselves the reason for disclosure, but they can increase the value of **awareness, inspection, preventive maintenance, and transparency.**

15. Dams Are Not Simply Liabilities

A discussion centered on failures, repair costs, and disclosure could easily leave the impression that dams are merely liabilities.

Historically, however, dams played an important role in **American industrialization.**

Before widespread steam power and electrification, small dams created hydraulic head for water wheels and turbines that powered:

- gristmills;
- sawmills;
- textile mills;
- iron works; and
- numerous other early industries.

Communities and manufacturing centers frequently developed around water-powered sites.

Many small dams remaining today are therefore physical remnants of an earlier generation of American infrastructure and industrial development.

Depending upon the site, dams and their impoundments can continue to provide:

- water supply and storage;
- irrigation;
- hydropower;
- recreation and fishing;
- scenic and landscape value;
- historical and cultural value;
- habitat associated with the impoundment; and
- other community benefits.

Water bodies may also provide some **localized microclimate moderation and evaporative cooling**, although the magnitude depends greatly on waterbody size, meteorology, vegetation, and surrounding land use, and evaporation also represents water loss.

There are also aesthetic values that conventional engineering benefit-cost calculations do not easily capture. Artificial waterfalls created by historic dams can provide attractive views, movement, and the sound of falling water.

As an engineer, I personally appreciate many of these features. That personal appreciation, however, should remain separate from the engineering decision concerning the future of a particular dam.

16. Maintain, Rehabilitate, Modify—or Remove?

Not every old dam should be preserved, but neither should removal automatically be regarded as the preferred solution.

Dams spanning significant streams can affect both upstream and downstream conditions, including:

- fish migration;
- sediment transport;
- stream connectivity;
- water temperature and quality;
- aquatic habitat;
- upstream and downstream water levels;
- flooding; and
- recreational safety.

At some sites, removing an obsolete dam can provide substantial ecological and safety benefits.

At others, rehabilitation, modification, fish ladders or fishways, bypass channels, or other measures may allow important dam benefits to remain while reducing particular environmental impacts.

Fish passage, however, should not be treated as universally equivalent to removal. A fishway may improve migration for particular species without restoring natural sediment transport, thermal conditions, channel processes, or complete ecological connectivity.

The decision is therefore broader than “**preserve or remove.**”

Maintain → rehabilitate → modify → improve fish passage → partially remove → fully remove

The appropriate choice depends on safety, environmental effects, historical significance, economics, water-management functions, recreation, aesthetics, and community priorities.

17. From Disclosure to Better Dam Stewardship

The NJ.com article began with a problem encountered by individual homeowners: people can acquire not only a house and land but also a piece of aging infrastructure carrying responsibilities they may not fully understand.

New Jersey’s new law is an important response.

But effective implementation can accomplish more than improving a real-estate transaction.

For **prospective buyers**, disclosure provides transparency.

For **current owners**, it can increase awareness of inspection, maintenance, repair, liability, and long-term responsibilities.

For **NJDEP and other government agencies**, implementation provides an opportunity to improve public guidance, information accessibility, and potentially the completeness of dam inventories.

For **engineers and researchers**, modern geospatial and AI technologies provide new opportunities to identify candidate structures and support better infrastructure information.

And for **policymakers**, the homeowner experiences highlighted by NJ.com raise a further question about how necessary safety improvements to privately owned dams should be financed when the benefits extend beyond the owner's property.

A dam can simultaneously be:

infrastructure + a source of risk + an environmental intervention + a historical artifact + a recreational or aesthetic resource.

Those characteristics vary enormously from one dam to another.

The appropriate approach is therefore neither to regard every old dam as a dangerous “money pit” nor to assume that every historic dam should remain indefinitely.

Disclosure provides transparency. Better information supports better inspection and maintenance. Earlier maintenance can reduce risks and potentially avoid much more expensive repairs. Better inventories can make the disclosure system itself more reliable. Together, these steps can move New Jersey from dam disclosure toward better long-term dam stewardship.

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- **Duda, J.J., et al., 2023** — “Patterns, drivers, and a predictive model of dam removal cost in the United States.” U.S. Geological Survey / *Frontiers in Ecology and Evolution*. The study compiled reported costs for 668 removed dams. <https://www.usgs.gov/publications/patterns-drivers-and-a-predictive-model-dam-removal-cost-united-states>
- **USGS — Compilation of cost estimates for dam removal projects in the United States — underlying empirical dataset.** <https://www.usgs.gov/data/compilation-cost-estimates-dam-removal-projects-united-states>
- **Guo, Q., July 2026** — technical brief on the Liulan Reservoir overtopping and local breach event, including extreme rainfall, gate-operation difficulties, overtopping, and breach development. <https://qizhong-guo.org/reflections/liulan-reservoir-overtopping-local-breach/>
- **Guo, Q., March 2026** — “Dam Disclosure” — earlier technical brief on dam disclosure, small and historic dams, homeowner responsibilities, field identification, and modern methods for screening possible dams and embankments. Originally prepared March 27, 2026; expanded March 29 and March 30, 2026. <https://qizhong-guo.org/reflections/dam-disclosure/>

Notes on Interpretation

Cost estimates: Except for published statistics specifically attributed to USGS or other sources, the cost ranges presented above are illustrative order-of-magnitude planning ranges based on general engineering experience. They are not NJDEP fee schedules, contractor quotations, or estimates for a particular property.

Inventory: The 1,728 figure represents the detailed April 2024 Class I–III database snapshot reported by the New Jersey State Auditor. A current authoritative statewide count of all Class IV dams has not been identified in the public materials reviewed for this reflection.

Professional assessment: The appropriate scope and professional requirements for inspecting or evaluating a particular dam depend on its classification, regulatory status, condition, and purpose. A homeowner’s visual observations can be useful but should not be treated as equivalent to a professional dam-safety inspection when one is required or warranted.